5.0 Logic

Logic is the study of reasoning and is used to determine whether statements are true or false.
In computer programming, logic is used to make decisions, compare values, and control how a program operates.

For example, a computer program might need to determine:

- Whether a password is correct.
- Whether a student has passed a course.
- Whether a number is positive or negative.
- Whether a user has permission to access a file.

Computers use logical conditions to decide which actions to perform.

5.1 Discuss with examples

5.1.1 Simple statement

A simple statement is a statement that expresses one idea and can be classified as either true or false, but not both. For example,

- 8 is greater than 5. (True)
- 12 is an odd number. (False)
- A keyboard is an input device. (True)
- 10 + 4 = 20. (False)

Simple statements are often represented by lowercase letters such as p, q, and r.

For example,

p: 10 is greater than 4.
q: 4 is an odd number.

p is true and q is false.

Questions and commands are not statements. “Enter your password” is a command. Subjective claims such as “Math is the hardest course.” do not have a clearly defined truth value without an agreed meaning.

5.1.2 Compound Statement

A compound statement is formed by combining two or more simple statements using logical operators.

The most common logical operators are:

Operator	Symbol	Name
AND	$\wedge$	Conjunction
OR	$\vee$	Disjunction
NOT	$\sim$	Negation

AND Operator

A compound statement using AND is true only when both statements are true.

Example:

p: A student submitted the assignment.

q: A student passed the test.

Compound statement:

$p \wedge q$: The student submitted the assignment AND passed the test.

This statement is true only if both conditions are true.

OR Operator

A compound statement using OR is true when at least one of the statements is true.

Example:

p: The user has a password.

q: The user has a security key.

Compound statement:

$p \vee q$: The user has a password OR a security key.

This statement is true if the user has a password, a security key, or both.

5.1.6 Negation

Negation reverses the truth value of a statement. If a statement is true, its negation is false. If a statement is false, its negation is true. Symbolically, the negation of p is ~p.

Statement	**Negation**
The computer is online.	The computer is not online.
$x > 5$	$x \leq 5$
$x = 8$	$x \neq 8$
All students passed the test.	At least one student did not pass the test.

Examples

1. Identify the following sentences as either True, False or Neither.

   a) Nova Scotia is Canada's most easterly province.
   b) What time is it?
   c) Christmas Day falls in December.
   d) This year has been brutal!

2. Consider the following simple statements:

p: It is cold.

q: It is windy.

Write each compound statement below using symbolic logic.

   a) It is cold and windy.
   b) It is not windy.
   c) It is not cold or it is windy.

d) It is not cold and it is not windy.

e) It is neither cold nor windy.

Note

Even though two statements may be written or worded differently, they can still be logically equivalent. One way to determine whether two statements are logically equivalent is to construct a truth table and compare their final columns.

5.2 Construct truth tables for logic statements

A truth table is a chart of rows and columns that lists every possible combination of truth values. For n simple statements, the number of rows is 2^n. One statement requires 2 rows, two require 4; three require 8, and so on.

Consider the simple statement, p. It has exactly two possible outcomes: True or False

A truth table for p appears as follows:

p
T
F

Now, let's consider two statements p and q. There are exactly four possible combinations (i.e. outcomes) when these two statements are considered together. Specifically, both could be true, both could be false, only p is true or only q is true.

Examples

Construct a truth table for the compound statements below.

a) $p \wedge q$

p	q	$p \wedge q$

b) $p \vee q$

p	q	$p \vee q$

c) $\sim p$

p	$\sim p$

We can also construct a truth table for more complex statements as the following problem will demonstrate.

Examples

Construct a truth table for each of the following.

a) $p \vee \sim q$

p	q		

b) $\sim p \wedge q$

p	q		

c) $\sim (\sim p \vee q)$

p	q			

d) $\sim (p \wedge \sim q)$

p	q			

One of the main uses of a truth table is to prove whether two compound statements are equivalent. This is accomplished by constructing a truth table for each and comparing the entries in the last column. If all entries match, the equivalence is proven. Otherwise, the statements are not equivalent.

Example

By constructing a truth table for each, show $\sim (p \wedge q)$ is not equivalent to $\sim p \wedge \sim q$.

